

CPARMS-WMF: A Clustering and Pairwise Association Rule Mining Signal for Weighted Matrix Factorization under Sparse Implicit Feedback

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Abstract

Weighted matrix factorization (WMF) struggles when interaction data are sparse. Side information can help, but user and item attributes are not always available. We examine whether patterns on the interaction matrix alone can help the limited history of an individual user.

We propose CPARMS-WMF, a feature-free framework that combines WMF with a Clustering and Pairwise Association Rule Mining Signal (CPARMS). CPARMS clusters users and items from positive feedback, mines single-antecedent rules over user-cluster, liked-item-cluster, and item-level tokens, and uses the resulting scores as an additional target. We evaluate CPARMS-WMF on five Amazon datasets with five random seeds and compare it with MostPop, Standard-WMF, and CoFactor-WMF.

CPARMS-WMF wins **73/75** baseline comparisons across dataset-seed runs: **25/25** against MostPop and **24/25** against each WMF baseline. On all five datasets, CPARMS-WMF achieves higher mean NDCG@10 than Standard-WMF (+**6.4%** to +**58.4%**), CoFactor-WMF (+**5.3%** to +**55.8%**), and MostPop (+**12.2%** to +**1724.4%**). For users with one positive interaction, it wins **21/25** against Standard-WMF and **19/25** against CoFactor-WMF, but only **13/25** against MostPop.

CPARMS-WMF improves over Standard-WMF and CoFactor-WMF. It also performs better in most comparisons for users with one positive interaction, but the popularity baseline remains competitive for this group.

Keywords: Recommender systems, Sparse implicit feedback, Weighted matrix factorization, Clustering, Association rule mining

1 Introduction

Recommender systems help people find relevant products and content in large catalogs, but recommendation quality often decreases for users with only one or a few recorded interactions.

Collaborative filtering, especially matrix factorization (MF) [1], remains widely used for its accuracy and scalability; it learns directly from interaction data, without requiring any side information. Weighted matrix factorization (WMF) [2] adapts MF to implicit feedback by assigning different confidence levels to observed and unobserved interactions. Alternating least squares (ALS) can optimize it efficiently. Both MF and WMF still needs enough interaction history: when a user’s history contains only a few interactions, the corresponding latent factors cannot be estimated reliably, and Top- K ranking quality drops; highly sparse datasets make this even worse [3].

One common solution to user sparsity is to add side information such as user profiles, demographic attributes, item descriptions, or cross-domain knowledge [4–6]. Side information can improve accuracy, but it is not always available. This gap motivates feature-free methods, which do not require user attributes or item metadata.

The interaction matrix contains patterns beyond the history of any single user or item. Clustering can reveal group-level preferences [7]; item co-occurrence can strengthen latent-factor learning [8]; and association rule mining can reveal directional relationships through support and confidence [9]. Previous work has combined item clustering with association rule mining to reduce sparsity [10], and CFCAI applies cluster-level rule mining to cold-start recommendation [11].

These studies motivate an integrated all clustering, association rule mining, and weighted matrix factorization in a single model. Clustering captures shared preference structure, association rules and find directional rules, and WMF uses this information for personalized ranking. CPARMS-WMF combines these complementary functions within a single objective.

We propose **CPARMS-WMF**, a feature-free framework for personalized recommendation under sparse implicit feedback. Its **Clustering and Pairwise Association Rule Mining Signal (CPARMS)** is derived only from the user–item interaction matrix. CPARMS clusters users and items by positive-interaction patterns, mines single-antecedent rules from user-cluster, liked-item-cluster, and item-level tokens, and converts those rules into a sparse user–item signal. An additional reconstruction term incorporates this signal into the ALS objective.

We evaluate only users and items observed in the fit window. Users and items that appear only in the evaluation window are excluded from training and the candidate

set, and their evaluation events are not scored. Among the remaining users, we include only those with at least one positive interaction in the fit window and at least one positive target in the evaluation window. Results are reported overall and separately for users with one, two, and three or more positive interactions.

CPARMS-WMF is compared with three feature-free baselines: a non-personalized popularity baseline (MostPop), Standard-WMF, and CoFactor-WMF.

The main contributions are:

1. We introduce CPARMS-WMF, a feature-free method for recommendation under sparse implicit feedback that not requires any user attributes or item metadata.
2. CPARMS combines user and item clustering with pairwise association rule mining. It mines user-cluster, liked-item-cluster, and item-level rules from the transactions.
3. We integrate the CPARMS signal into WMF through an additional reconstruction term optimized by alternating least squares.
4. We evaluate CPARMS-WMF on five Amazon domains using five random seeds. NDCG@10 is the primary metric, with NDCG@20, NDCG@50, NDCG@100, and NDCG@200 reported as secondary measures. The analysis covers mean performance and pairwise dataset-seed comparisons against all three baselines, both overall and for users with one, two, and three or more positive interactions. It also identifies where CPARMS-WMF improves over the WMF baselines and where popularity remains competitive.

2 Related Work

We first review matrix factorization for implicit feedback and then consider feature-free methods that extract additional structure from the interaction matrix.

2.1 Matrix Factorization and Weighted Matrix Factorization

Matrix factorization (MF) is a core recommendation technique. It places users and items in a shared low-dimensional latent space and estimates preferences from how their latent factors interact [1].

Weighted matrix factorization (WMF) extends MF to implicit-feedback data by modeling binary preference alongside confidence weights. WMF learns its latent factors by minimizing a confidence-weighted reconstruction loss, most often with alternating least squares [2].

Both methods remain sensitive to sparsity: when observed interactions are few, user representations rely on limited information, estimates become unreliable, and Top- K ranking quality drops [3]. This problem motivates additional signals derived from interactions.

2.2 Non-personalized Recommendation

Ranking candidate items by global popularity, identically for every user, is the standard non-personalized reference point in offline recommender evaluation [12, 13]. Because this baseline does not use individual user histories, a personalized model

that cannot outperform it has not demonstrated an accuracy benefit from personalization under this evaluation protocol. Under strong sparsity, popularity remains a strong baseline because items that are popular in the fit window are likely to reappear in a future evaluation window [14], and Section 5 shows that it outperforms Standard-WMF on two of our five datasets.

2.3 Hybrid Matrix Factorization with Side Information

A common approach to reducing sparsity is to use side information with collaborative filtering. Content-boosted matrix factorization adds item attributes directly to the factorization, improving predictive accuracy and making recommendations easier to interpret [4].

Cross-domain matrix factorization transfers information from related domains by using item metadata and semantic relationships as links between source and target items. It is especially helpful for new users [5]. More broadly, hybrid recommender systems combine collaborative and content-based filtering so that each approach can address the weaknesses of the other [6].

2.4 Hybrid Matrix Factorization without Side Information

Unlike hybrid methods that rely on external attributes, feature-free methods draw additional signals only from the user-item interaction matrix. Clustering-based approaches identify preference structures shared by users or items [7], while association-rule approaches mine directional item relationships from transaction data [9]. Some studies combine two techniques or connect one of them to matrix factorization. Clustering is used to reduce the dimensionality of the item space, followed by association rule mining [10]. CFCAI mines rules between clusters to support cold-start recommendation [11]. CoFactor instead adds an item-item co-occurrence term to the factorization objective, supplying information for items with few observed interactions [8]. To our knowledge, none of the studies reviewed above integrates clustering, association rule mining, and weighted matrix factorization within one model.

CoFactor is a relevant matrix-factorization baseline because it links implicit-feedback WMF with an item-item co-occurrence signal derived from the same interaction matrix [8]. The original model jointly reconstructs the user-item interaction matrix and an item-item shifted positive pointwise mutual information (SPPMI) matrix. Using the notation of this paper, a CoFactor-style objective is

$$\begin{aligned} \mathcal{L}_{\text{CoFactor}} = & \sum_{u \in \mathcal{U}} \sum_{i \in \mathcal{I}} c_{ui} (\pi_{ui} - \mathbf{p}_u^\top \mathbf{q}_i)^2 \\ & + \gamma_{\text{co}} \sum_{(i,j) \in \mathcal{M}} (m_{ij} - \mathbf{q}_i^\top \mathbf{z}_j - b_i - d_j)^2 \\ & + \lambda (\|\mathbf{P}\|_F^2 + \|\mathbf{Q}\|_F^2) + \lambda_{\text{ctx}} \|\mathbf{Z}\|_F^2. \end{aligned} \quad (1)$$

Here π_{ui} is the binary implicit preference, c_{ui} is its WMF confidence weight, $\mathbf{p}_u$ and $\mathbf{q}_i$ are user and item factors, m_{ij} is the SPPMI value for item i and context item

j , $\mathcal{M}$ indexes nonzero SPPMI entries, $\mathbf{z}_j$ is a context factor, b_i and d_j are item-side and context-side biases, γ_{co} controls the co-occurrence reconstruction weight, and λ_{ctx} regularizes the context factors. The shared item factor $\mathbf{q}_i$ connects the WMF component to the SPPMI reconstruction component. We compute the SPPMI entries from item co-occurrence counts c_{ij} , where c_{ij} is the number of users who positively interacted with both items i and j , and we set the diagonal to zero:

$$\text{SPPMI}_{ij} = \max \left(\log \frac{c_{ij}N}{f_i f_j} - \log k, 0 \right), \quad (2)$$

where N is the total co-occurrence count over nonzero entries, f_i and f_j are the marginal co-occurrence counts of items i and j , and k is the number of negative samples, whose logarithm shifts the pointwise mutual information. CoFactor therefore adds a signal derived from interactions, but this signal encodes item-level co-occurrence rather than group-level user preference patterns.

Equation (1) keeps the co-occurrence term weighted and the WMF term at unit scale. The original formulation instead applies a relative scale ℓ to the WMF term and its user and item regularizers. Dividing the objective by ℓ gives Eq. (1) with $\gamma_{\text{co}} = 1/\ell$ and $\lambda_{\text{ctx}} = \gamma_{\text{co}}\lambda$. Our implementation and hyperparameter search keep this relationship: we sample ℓ and derive γ_{co} and λ_{ctx} from it (Section 4.5).

2.5 How the Proposed Method Differs

The feature-free methods reviewed above use clustering, association rule mining, and weighted matrix factorization mostly separately. CPARMS-WMF combines all three in one framework: clustering provides group-level preference structure shared across the members of each user or item cluster, pairwise association rules provide directional confidence-weighted information, and WMF provides personalized latent-factor ranking. CoFactor-WMF adds SPPMI item co-occurrence to matrix factorization, which makes it a suitable baseline for CPARMS-WMF because both use extra information from the interaction matrix. The main difference is that CoFactor-WMF regularizes shared item embeddings with item-level co-occurrence, while CPARMS builds a user-item signal from item-item, user-cluster-item, and liked-item-cluster-item relations.

CPARMS uses three token families: user-cluster, liked-item-cluster, and item-level tokens. User-cluster tokens provide a generic signal shared by users in the same cluster, while the other two provide personalized signals based on each user’s positive history.

This comparison raises two questions. First, can CPARMS-WMF improve personalized ranking under sparse implicit feedback, beating Standard-WMF, CoFactor-WMF, and the non-personalized popularity baseline? Second, does this improvement remain for users with extremely short histories, including when compared with the popularity baseline? The results in Section 5 support the first claim but only partly support the second.

3 Proposed Method

CPARMS-WMF is designed to improve Top- K recommendation under sparse implicit feedback by deriving its CPARMS signal through clustering and pairwise association rule mining on the user-item interaction matrix.

Three main components make up the framework: (i) interaction-based clustering of users and items, followed by pairwise association rule mining; (ii) construction of a sparse user-item recommendation signal from the rules; and (iii) integration of that signal into a weighted matrix-factorization objective optimized by alternating least squares.

3.1 Notation and Feedback Views

Let $\mathcal{U} = \{1, \dots, U\}$ and $\mathcal{I} = \{1, \dots, I\}$ be the user and item index sets, and let $\mathbf{Y} = [y_{ui}] \in \mathbb{R}_{\geq 0}^{U \times I}$ be the interaction matrix of the fit window. From $\mathbf{Y}$ the pipeline derives two binary views and keeps them strictly separate:

$$\underbrace{b_{ui} = \mathbb{I}[y_{ui} > 0]}_{\mathbf{B} \text{ (seen)}}, \quad \underbrace{\ell_{ui} = \mathbb{I}[y_{ui} > \theta]}_{\mathbf{L} \text{ (liked)}}, \quad \theta = 3. \quad (3)$$

$\mathbf{L}$ is the positive feedback (“liked”) view: at $\theta = 3$, integer ratings of four or higher count as positive. $\mathbf{B}$ is the “seen” view, recording every observed rating at any level.

All learned inputs are derived from $\mathbf{L}$: the cluster assignments, transaction tokens, association rules, SPPMI matrix, and WMF preference target. The matrix $\mathbf{B}$ is used only to mask candidates. An item is therefore treated as seen after any rating, even one below the like threshold, and is not recommended again. For user u , let $\Omega_u^+ = \{i \mid \ell_{ui} = 1\}$ denote the positive history, $\Omega_u = \{i \mid b_{ui} = 1\} \supseteq \Omega_u^+$ the seen set, and $n_u = |\Omega_u^+|$.

3.2 CPARMS: Clustering and Pairwise Association Rule Construction

Previous studies show that combining clustering with association rule mining can improve collaborative filtering on sparse data [7, 9, 10]. CPARMS follows this approach and builds an additional sparse signal using only positive interactions.

We cluster users and items separately with K-means [15] on the binary positive feedback matrix. K-means is fitted only to active rows, which contain at least one positive interaction, and the fitted model then assigns every row—including an all-zero row—to its nearest cluster. Excluding all-zero rows from fitting prevents identical inactive rows from forming an uninformative cluster. User clustering operates on the rows of $\mathbf{L}$ and produces K_U clusters; the one-hot matrix $\mathbf{U} \in \{0, 1\}^{U \times K_U}$ records each user’s assignment. Item clustering operates on the rows of $\mathbf{L}^\top$ and produces the one-hot assignment matrix $\mathbf{G} \in \{0, 1\}^{I \times K_I}$. We then construct a binary liked-item-cluster matrix that records whether a user’s positive history contains any item from each cluster:

$$\mathbf{E} = \mathbb{I}[\mathbf{L}\mathbf{G} > 0] \in \{0, 1\}^{U \times K_I}. \quad (4)$$

The search grid contains only positive values for both cluster counts, so both cluster-token families appear in every CPARMS-WMF configuration by construction. When $K_U = 1$, all users share the same user-cluster token, which acts as a dataset-wide popularity signal. Validation selects this setting for some datasets (Section 5).

For each user, we stack three token families (user-cluster tokens, liked-item-cluster tokens, and item-level tokens) into a single transaction matrix:

$$\mathbf{T} = [\mathbf{L} \mid \mathbf{U} \mid \mathbf{E}] \in \{0, 1\}^{U \times M}, \quad M = I + K_U + K_I. \quad (5)$$

Let $\mathcal{A}$ index all M tokens, and let $\mathcal{R}$ be the item-level tokens, which are the only valid consequents. We use user-cluster and liked-item-cluster tokens only as antecedents, so every rule has an individual item as its consequent.

We mine single-antecedent directed rules from joint token counts. Given the pair-count matrix $\mathbf{C} = \mathbf{T}^\top \mathbf{T}$ and token frequency $n_t = \sum_u T_{ut}$, the metrics for a rule $a \rightarrow b$ are:

$$\text{sup}(a \rightarrow b) = \frac{C_{ab}}{U}, \quad \text{conf}(a \rightarrow b) = \frac{C_{ab}}{n_a}, \quad \text{lift}(a \rightarrow b) = \frac{U C_{ab}}{n_a n_b}. \quad (6)$$

Confidence is the conditional probability that a user linked to antecedent a is also linked to consequent b , and therefore measures the rule’s directional reliability. Lift compares the joint frequency with that expected under independence: $\text{lift} > 1$ indicates a positive association beyond chance, whereas $\text{lift} \approx 1$ indicates a relationship already explained by the consequent’s overall popularity. We consider only directional pairs with $a \neq b$ and $b \in \mathcal{R}$. A rule is selected when it meets the configured minimum support, confidence, and lift. All three thresholds are tuned on validation data (Section 4.5). The rule-selection indicator is

$$D_{ab} = \mathbb{I}[\text{sup}(a \rightarrow b) \geq \sigma_{\min} \wedge \text{conf}(a \rightarrow b) \geq \rho_{\min} \wedge \text{lift}(a \rightarrow b) \geq \ell_{\min}]. \quad (7)$$

Each selected rule keeps its confidence as its weight:

$$W_{ab} = D_{ab} \text{conf}(a \rightarrow b). \quad (8)$$

Thus, selected rules carry weight in proportion to their directional reliability, while support and lift serve only as noise filters through Eq. (7). Collecting the selected rules produces $\mathbf{W} \in \mathbb{R}^{M \times I}$, which encodes the filtered relationships from antecedent tokens to individual-item consequents.

3.3 CPARMS Signal Generation

Figure 1 summarizes the implemented pipeline. CPARMS produces two sources of information. Rules triggered by user-cluster tokens in $\mathbf{U}$ give a generic score shared by members of the same cluster. Rules triggered by liked-item-cluster tokens in $\mathbf{E}$ or item-level tokens in $\mathbf{L}$ give personalized scores because both token types depend on the user’s own positive history.

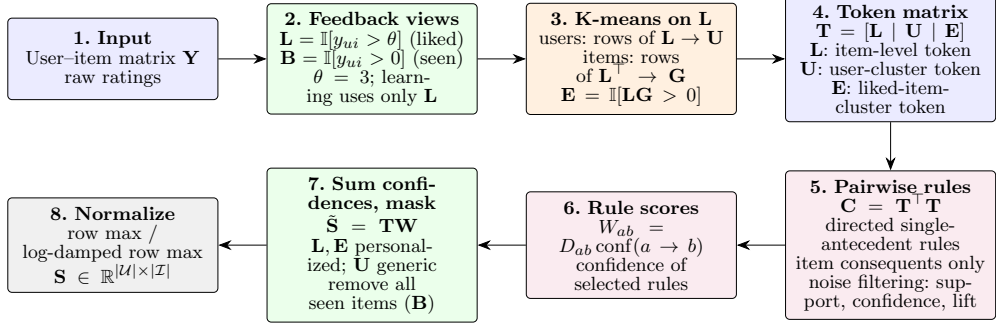


Fig. 1 Overview of the CPARMS signal-generation pipeline, from the raw user–item interaction matrix $\mathbf{Y}$ to the normalized user–item signal matrix $\mathbf{S}$. All learning happens on the liked view $\mathbf{L}$; the seen view $\mathbf{B}$ enters only as the candidate mask in stage 7

To form the raw signal, CPARMS applies the rule scores in $\mathbf{W}$ to every token in a user’s transaction row, summing, for each candidate item, the scores of all rules triggered by the user’s active tokens:

$$\tilde{\mathbf{S}} = \mathbf{T}\mathbf{W} \in \mathbb{R}^{U \times I}. \quad (9)$$

Because the token families occupy separate column blocks of $\mathbf{T}$, this sum divides exactly into the two types of information. We define

$$\mathbf{T}^{(p)} = [\mathbf{L} \mid \mathbf{0}_{U \times K_U} \mid \mathbf{E}], \quad \mathbf{T}^{(g)} = [\mathbf{0}_{U \times I} \mid \mathbf{U} \mid \mathbf{0}_{U \times K_I}]. \quad (10)$$

The raw signal is the equal-weight sum of the personalized and generic signals:

$$\tilde{\mathbf{S}} = \mathbf{T}^{(p)}\mathbf{W} + \mathbf{T}^{(g)}\mathbf{W} = \tilde{\mathbf{S}}^{(p)} + \tilde{\mathbf{S}}^{(g)}. \quad (11)$$

There is no blending parameter; the way the two signals combine depends entirely on token activation. Every user activates one user-cluster token, so the generic signal gives a shared cluster-level score vector to each cluster member. In contrast, the number of active personalized antecedents grows with n_u , so the personalized signal becomes stronger as positive history grows.

Equation (11) shows that a user with $n_u = 0$ would receive only the generic signal. We do not evaluate this case because users with no positive interactions are outside the evaluation population (Section 4.2).

After summing the confidence scores, we set the score of every seen item to zero using $\mathbf{B}$, so the signal applies only to unseen items. Since the seen set includes all positive items, none of them retains a signal score.

The unseen signal is then normalized using one of two searched modes. Row-wise maximum scaling divides each user row by its own maximum. log-transformed row-wise scaling first compresses heavy-tailed rule-score sums with $\log(1 + \cdot)$ so that a single dominant item does not compress the remaining scores in the row toward zero.

Let $\tilde{s}_{ui}$ be the unseen raw signal. The two modes are:

$$s_{ui} = \frac{\tilde{s}_{ui}}{\max_j \tilde{s}_{uj}}, \quad \text{or} \quad s_{ui} = \frac{\log(1 + \tilde{s}_{ui})}{\max_j \log(1 + \tilde{s}_{uj})}. \quad (12)$$

Empty rows remain zero, and $0 \leq s_{ui} \leq 1$ under either mode. This yields the sparse, non-negative CPARMS signal matrix $\mathbf{S} = [s_{ui}] \in \mathbb{R}_{\geq 0}^{U \times I}$. We write $\Psi_u = \{i \mid s_{ui} > 0\}$ for each user's set of signal-supported items; by construction, $\Psi_u \cap \Omega_u = \emptyset$.

3.4 CPARMS-WMF: Signal-Regularized Weighted Matrix Factorization

CPARMS-WMF integrates the signal into a single matrix-factorization objective. It adds a reconstruction term over signal-supported items to implicit-feedback WMF [2], similar to the additional reconstruction regularizer in CoFactor [8]. We compute interaction scores as the inner product of latent factors:

$$\hat{y}_{ui} = \mathbf{p}_u^\top \mathbf{q}_i, \quad \mathbf{p}_u, \mathbf{q}_i \in \mathbb{R}^K, \quad (13)$$

where K is the latent dimension. Stacking row-wise gives $\mathbf{P} \in \mathbb{R}^{U \times K}$ and $\mathbf{Q} \in \mathbb{R}^{I \times K}$. The preference target is the liked view of Eq. (3), and every pair gets an implicit-feedback confidence that up-weights positives,

$$\pi_{ui} = \ell_{ui}, \quad c_{ui} = 1 + \alpha \pi_{ui}, \quad (14)$$

where $\alpha \geq 0$ sets the extra confidence placed on positive interactions. One threshold $\theta = 3$ defines positives across CPARMS construction, SPPMI construction, WMF training, and evaluation relevance, following the explicit-to-implicit preprocessing of CoFactor [8].

The training objective combines a confidence-weighted reconstruction term over all user-item pairs, a γ -weighted reconstruction term over signal-supported pairs, and an ℓ_2 penalty,

$$\begin{aligned} \mathcal{L} = & \sum_{u \in \mathcal{U}} \sum_{i \in \mathcal{I}} c_{ui} (\pi_{ui} - \hat{y}_{ui})^2 + \gamma \sum_{u \in \mathcal{U}} \sum_{i \in \Psi_u} (s_{ui} - \hat{y}_{ui})^2 \\ & + \lambda (\|\mathbf{P}\|_F^2 + \|\mathbf{Q}\|_F^2). \end{aligned} \quad (15)$$

The first term is the standard implicit WMF loss. It pulls predictions toward 1 on observed positives with high confidence $1 + \alpha$ and toward 0 on all remaining pairs with unit confidence. The second term aligns predictions with the CPARMS signal on signal-supported candidate items. Every signal-supported item lies outside the seen set ($\Psi_u \cap \Omega_u = \emptyset$), so the signal never competes with an observed positive target. For each item $i \in \Psi_u$, the implicit term pulls the prediction toward 0 with unit confidence, while the γ term pulls it toward s_{ui} . Together, these terms raise CPARMS-supported

candidates above other unobserved items. The second term therefore provides additional information: each signal-supported entry receives a nonzero target outside the user’s observed history, whereas the implicit term alone would treat it only as an unobserved zero. Users with short histories gain extra least-squares constraints wherever the rules support unseen candidates. The hyperparameter γ sets the strength of the signal regularizer, and λ is the ℓ_2 coefficient.

3.5 Optimization via Alternating Least Squares

Equation (15) is a weighted least-squares objective. It is quadratic in $\mathbf{P}$ when $\mathbf{Q}$ is fixed and vice versa. For this reason, we optimize CPARMS-WMF with ALS, which allows a closed-form update for each factor row [2]. Let $\mathbf{C}^u = \text{diag}(c_{u1}, \dots, c_{uI})$ be the $I \times I$ confidence diagonal, and let $\boldsymbol{\pi}_u = (\pi_{u1}, \dots, \pi_{uI})^\top$ be the preference vector for user u . Minimizing Eq. (15) over $\mathbf{p}_u$ gives the normal equation:

$$\left(\mathbf{Q}^\top \mathbf{C}^u \mathbf{Q} + \gamma \sum_{i \in \Psi_u} \mathbf{q}_i \mathbf{q}_i^\top + \lambda \mathbf{I}_K \right) \mathbf{p}_u = \mathbf{Q}^\top \mathbf{C}^u \boldsymbol{\pi}_u + \gamma \sum_{i \in \Psi_u} s_{ui} \mathbf{q}_i. \quad (16)$$

Following the standard implicit-feedback decomposition, we expand the confidence-weighted terms around the unit-confidence baseline. As a result, we only need to sum the contributions from positive and signal-supported items for each user:

$$\begin{aligned} \mathbf{A}_u &= \mathbf{Q}^\top \mathbf{Q} + \lambda \mathbf{I}_K + \alpha \sum_{i \in \Omega_u^+} \mathbf{q}_i \mathbf{q}_i^\top + \gamma \sum_{i \in \Psi_u} \mathbf{q}_i \mathbf{q}_i^\top, \\ \mathbf{d}_u &= (1 + \alpha) \sum_{i \in \Omega_u^+} \mathbf{q}_i + \gamma \sum_{i \in \Psi_u} s_{ui} \mathbf{q}_i, \end{aligned} \quad (17)$$

where $\mathbf{A}_u \in \mathbb{R}^{K \times K}$ is the user-specific system matrix and $\mathbf{d}_u \in \mathbb{R}^K$ its right-hand side. The user factor follows from solving

$$\mathbf{A}_u \mathbf{p}_u = \mathbf{d}_u. \quad (18)$$

We precompute the shared term $\mathbf{Q}^\top \mathbf{Q}$ once per sweep, so each user update costs only the rank-one contributions of that user’s positive and signal-supported items. Item-factor updates follow the same procedure with the roles of users and items swapped. We initialize both factor matrices with small-variance random values drawn from a fixed seed, and the algorithm then alternates user and item sweeps for a fixed number of iterations. At inference, we mask all seen items and rank the remaining items for user u by $\hat{y}_{ui} = \mathbf{p}_u^\top \mathbf{q}_i$.

4 Experimental Setup

The experiments use five datasets, a global temporal split, three baselines, and NDCG-based model selection and evaluation.

4.1 Datasets

We use five public benchmark datasets from the 2018 Amazon review corpus [16]: Luxury Beauty, Industrial and Scientific, Prime Pantry, Digital Music, and Musical Instruments. The released 5-core datasets provide the input, and we use only the interaction fields. We then de-duplicate all datasets, keeping the earliest review or, when several reviews share the earliest timestamp, the one with the highest rating. Table 1 reports the resulting statistics.

Table 1 Statistics of the Amazon benchmark datasets after de-duplication

Dataset	#Users	#Items	#Interactions	Int./User	Avg. Rating	Sparsity (%)
Luxury Beauty	3,819	1,581	28,074	7.35	4.24	99.54
Industrial & Scientific	11,041	5,334	72,025	6.52	4.52	99.88
Pantry	14,180	4,970	131,114	9.25	4.54	99.81
Music	16,566	11,797	145,292	8.77	4.70	99.93
Instruments	27,530	10,620	219,156	7.96	4.47	99.93

The published datasets are 5-core based on raw review counts, but our pipeline removes repeated reviews of the same item by the same user, and this step does not preserve the 5-core property. After de-duplication, many users have fewer than five distinct items; in Luxury Beauty, for example, 1,188 of 3,819 users do. For this reason, we describe these datasets as derived from the 5-core releases rather than strictly 5-core, and they are sparser than the 5-core label suggests. Sparsity exceeds 99.5% in every domain. This high sparsity makes the datasets difficult for feature-free methods.

4.2 Data Splitting Protocol

We use a global temporal split to preserve chronological order and avoid training on future interactions [17]. The procedure sorts all events from oldest to newest, using stable input-row order to break timestamp ties. Row position defines an approximate 70/10/20 training, validation, and test split. Each boundary is then moved to the nearest change in timestamp so that events recorded at the same time remain in one partition.

The fit window alone defines which users and items the model can use:

- Tuning stage: fit on training, evaluate on validation.
- Final stage: fit on training-plus-validation, evaluate on test.

Users and items that appear only in a future window never enter the fit matrix as empty rows or columns, so they cannot affect training or appear as candidate items. In particular, they cannot change the U denominator of rule support and lift in Eq. (6). We drop rather than score evaluation events involving unknown users or items. Hyperparameter tuning uses the training and validation sets, after which we retrain the selected configuration on the combined training-plus-validation interactions before the held-out test evaluation.

We divide evaluation users into three positive interaction groups according to the size of their positive history n_u : 1, 2, and 3^+ interactions. An evaluated user must have $n_u \geq 1$ and at least one positive target in the evaluation window. Users with $n_u = 0$ are excluded because an all-zero positive feedback row gives both WMF baselines a zero user factor when solved with ALS. Scoring CPARMS’s cluster-level information against baselines with no user-specific estimate would answer a different question; we leave a fair evaluation of that case for future work.

Under this protocol, the test evaluation covers 1,019 users in Luxury Beauty, 3,904 in Industrial & Scientific, 2,722 in Pantry, 4,080 in Music, and 8,541 in Instruments. Among these users, 110, 456, 343, 462, and 1,017, respectively, have only one positive interaction in the fit window. These evaluated users represent 26.7%, 35.4%, 19.2%, 24.6%, and 31.0% of the total users in each dataset, respectively. Most evaluated users have at least three positive interactions and therefore belong to the 3^+ group.

4.3 Baseline Models

We compare CPARMS-WMF with three feature-free baselines. Every model uses the same temporal splits, relevance threshold $\theta = 3$, metric for model selection, and final test protocol.

MostPop ranks candidate items by global popularity, defined as the number of users who positively interacted with each item in the fit window, and applies the same ranking to every user [12]. MostPop is non-personalized, deterministic, and has no hyperparameters. As a result, its scores stay the same across seeds, and its standard deviation is exactly zero. We include it as a reference point to test whether personalization improves accuracy beyond popularity.

Standard-WMF is the implicit-feedback WMF model of Hu et al. [2]. It uses binary preference $\pi_{ui} = \ell_{ui}$ and confidence $c_{ui} = 1 + \alpha\pi_{ui}$ and learns user and item factors through ℓ_2 regularization and ALS.

CoFactor-WMF implements the model in Section 2. It shares the WMF item factors with a reconstruction of the item–item SPPMI matrix in Eq. (2). The model uses unit weights on nonzero SPPMI entries, separate context factors, item and context biases, and ALS updates for user, item, context, and bias parameters [8].

The search uses CoFactor’s original parameters. It samples the relative WMF scale ℓ and derives $\gamma_{co} = 1/\ell$ and $\lambda_{ctx} = \gamma_{co}\lambda$ instead of treating the co-occurrence weight and context regularizer as independent parameters. Both search grids come directly from the original paper: $\ell \in \{0.01, 0.05, 0.1, 0.5, 1, 5, 10\}$ and the number of negative samples $k \in \{1, 2, 5, 10, 50\}$ [8]. Sampling these quantities independently could assign negligible weight to the co-occurrence term, effectively reducing CoFactor-WMF toward Standard-WMF. We therefore preserve the coupling used in the original parameterization.

4.4 Evaluation Metrics

We measure recommendation performance with normalized discounted cumulative gain (NDCG) at cutoffs 10, 20, 50, 100, and 200, following standard offline evaluation practice [13]. Relevance is binary: a held-out interaction counts as relevant once its

rating exceeds $\theta = 3$. For user u , let $\text{rel}_{u,r} = 1$ when the item ranked at position r is relevant and 0 otherwise, and let $\text{IDCG}@K_u$ denote the ideal discounted cumulative gain, that is, the maximum discounted cumulative gain attainable for that user’s known relevant test items at cutoff K . Then

$$\text{NDCG}@K_u = \frac{\sum_{r=1}^K \frac{\text{rel}_{u,r}}{\log_2(r+1)}}{\text{IDCG}@K_u}. \quad (19)$$

For each user, we remove all seen items ($\mathbf{B}$, not only liked items) from the candidate ranking and break score ties by item index. Within each positive interaction group, scores are averaged over the users who meet the criteria in Section 4.2. We report the mean and sample standard deviation over seeds 42–46; all NDCG values are percentages. NDCG@10 is the primary selection metric, while larger cutoffs measure ranking quality over broader candidate lists.

4.5 Hyperparameter Search

We choose model hyperparameters by randomized search [18]. For each seed, we sample 30 aligned configurations once and reuse them across all datasets. Within a trial, the WMF-based models share the factorization hyperparameters (K , λ , ALS sweeps, and α). We draw CPARMS-specific choices from the same random stream after these shared values, while CoFactor-specific choices use an offset random stream. For each dataset–seed pair, we keep the configuration with the best validation NDCG@10, then retrain the model on the combined training-plus-validation set before testing.

Table 2 summarizes the search space. The CPARMS rule thresholds include zero and small positive values because rule-metric scales vary greatly across antecedent families under extreme sparsity. We keep the two reconstruction weights separate: CPARMS’s γ weights a user–item signal on a $[0, 1]$ scale, while CoFactor’s γ_{co} weights an SPPMI target whose scale depends on co-occurrence statistics.

5 Results

We first report overall NDCG and then analyze performance across positive-interaction groups. All NDCG values are percentages reported as means over seeds 42–46, with sample standard deviations where applicable. The best values appear in bold; second-best values are underlined. Pairwise seed wins also show whether the advantage appears in many runs or only a few.

5.1 Overall Improvement in Personalized Ranking

The relative improvement of CPARMS-WMF over baseline b is

$$\Delta_b = \frac{\text{NDCG}_{\text{CPARMS-WMF}} - \text{NDCG}_b}{\text{NDCG}_b} \times 100. \quad (20)$$

Table 2 Hyperparameter search space and descriptions. The search samples thirty configurations per seed and evaluates them on every dataset

Hyperparameter	Search Space	Applies to	Description
Latent dimension K	{10, 20, 30, 40}	all WMF	latent factors
Regularization λ	{0.0001, 0.001, 0.01, 0.1}	all WMF	ℓ_2 penalty
ALS sweeps	{15, 20, 25, 30}	all WMF	ALS iterations
Confidence α	{10, 20, 40, 80}	all WMF	confidence on positives
Signal weight γ	{0.5, 1.0, 2.0, 4.0}	CPARMS	signal-reconstruction weight
Signal normalization	{row_max, log_row_max}	CPARMS	signal scaling mode
User clusters K_U	{1, 2, 3, 5}	CPARMS	user clusters
Item clusters K_I	{1, 2, 3, 5}	CPARMS	item clusters
Min. support $\sigma_{\min}$	{0, 0.00005, 0.0001, 0.0003}	CPARMS	support noise filter
Min. confidence $\rho_{\min}$	{0, 0.0005, 0.001, 0.002}	CPARMS	confidence noise filter
Min. lift $\ell_{\min}$	{0, 0.2, 0.5, 0.8}	CPARMS	lift noise filter
Relative WMF scale ℓ	{0.01, 0.05, 0.1, 0.5, 1, 5, 10}	CoFactor	sets $\gamma_{\text{co}} = 1/\ell$
SPPMI shift k	{1, 2, 5, 10, 50}	CoFactor	negative samples
Context regularization	derived: $\lambda_{\text{ctx}} = \gamma_{\text{co}}\lambda$	CoFactor	not searched (derived)

5.1.1 Performance against the WMF baselines

At NDCG@10, CPARMS-WMF wins 73 of the 75 dataset–seed comparisons against the three baselines (Table 3). Against the two WMF baselines alone, it wins 24/25 comparisons each, has the best overall mean on all five datasets, and improves over Standard-WMF by +6.4% (Music), +7.3% (Pantry), +8.9% (Luxury Beauty), +32.7% (Instruments), and +58.4% (Industrial & Scientific). The gains over CoFactor-WMF are +5.3% (Music), +7.4% (Luxury Beauty), +10.6% (Pantry), +27.6% (Instruments), and +55.8% (Industrial & Scientific). Its only losses occur on Pantry seed 45 against both WMF baselines. Table 4 provides the corresponding breakdown by positive interaction group. The advantage is not limited to the selection cutoff: Table 5 shows that CPARMS-WMF has the best overall mean at every cutoff from 10 to 200 on all five datasets.

5.1.2 Performance against popularity

MostPop remains an important reference because it beats both WMF baselines overall on Industrial & Scientific and Instruments. Even so, CPARMS-WMF beats MostPop in all 25 overall dataset–seed runs and has a higher mean on every dataset. Its improvement ranges from +12.2% on Industrial & Scientific to +1724.4% on Luxury Beauty. MostPop’s near-zero NDCG@10 (0.25%) causes the very large percentage on Luxury Beauty, where the popularity baseline is weakest. When we weight the five dataset means equally, CPARMS-WMF reaches 2.79%, compared with 2.32% for Standard-WMF, 2.35% for CoFactor-WMF, and 1.36% for MostPop.

5.2 Positive Interaction Groups

Table 4 divides users by the number of positive interactions. For users with one positive interaction, CPARMS-WMF beats Standard-WMF in 21/25 runs and CoFactor-WMF in 19/25. Its largest mean gains over Standard-WMF occur on Industrial & Scientific (+142.6%) and Instruments (+68.2%). The gains are not uniform, however. On Luxury

Table 3 Pairwise NDCG@10 wins by CPARMS-WMF across the 25 dataset–seed runs. Each cell reports wins out of 25 using unrounded scores. The one-interaction row covers users with exactly one positive interaction

Positive interaction group	vs. MostPop	vs. Standard-WMF	vs. CoFactor-WMF
All users	25	24	24
1 interaction	13	21	19
2 interactions	18	22	23
3+ interactions	25	23	24

Beauty, the mean for users with one positive interaction is lower than both WMF baselines (9.24 versus 9.53 and 9.79), with only 2/5 seed wins against each; on Music, it is lower than CoFactor-WMF. Results against MostPop are mixed: CPARMS-WMF wins 13/25 runs for users with one positive interaction and has the best mean for these users only on Pantry. The evidence therefore supports an improvement over the two WMF baselines in most runs for users with one positive interaction, but not on every dataset and not consistently over popularity.

At two interactions, CPARMS-WMF wins 22/25 against Standard-WMF, 23/25 against CoFactor-WMF, and 18/25 against MostPop, and has the best mean on three datasets; the two remaining best means go to MostPop on Industrial & Scientific and CoFactor-WMF on Pantry. At three or more interactions, it has the best mean on all five datasets and wins 25/25 against MostPop, 23/25 against Standard-WMF, and 24/25 against CoFactor-WMF. This pattern fits the method design: item-level and liked-item-cluster antecedents provide more user-specific information as n_u grows.

5.3 Computational Cost

Table 6 reports training time. CPARMS-WMF costs more to train than either WMF baseline: total time ranges from 0.334 minutes on Luxury Beauty to 8.818 minutes on Instruments. This is 3.1–14.2× the training time of Standard-WMF and 2.1–6.6× that of CoFactor-WMF. Most of the extra time comes from denser signal-regularized ALS updates rather than signal construction.

5.4 Hyperparameter Behavior

Because we select a configuration for each dataset–seed pair using validation NDCG@10, the chosen CPARMS-WMF configurations vary. Table 7 lists these configurations across the 25 selections:

1. **Cluster counts vary, and the single-cluster setting is common.** Validation selects $K_U = 1, 2, 3, 5$ in 9, 2, 8, 6 runs, respectively, and $K_I = 1, 2, 3, 5$ in 6, 7, 7, 5 runs, respectively. The search space never disables either token family, so the presence of both families is part of the design rather than a finding. The common choice $K_U = 1$ is a special case: every user shares one cluster token, so its rules become a single dataset-wide score vector proportional to the number of users who positively interacted with each item (Eq. (6)). This vector is a popularity-like prior

rather than group-specific information. The setting appears in all five Industrial & Scientific runs, where popularity is strongest, while the other datasets mostly use $K_U > 1$.

2. **Rule filtering varies across a broad grid.** Zero thresholds appear 9, 7, and 8 times for support, confidence, and lift, respectively. The most common positive support value is 0.0001 (10 selections). The confidence thresholds 0.0005 and 0.001 each appear 8 times, while the selected lift thresholds span the full grid, with the upper values 0.5 and 0.8 chosen 13 times in total. All searched thresholds are low in absolute terms; even the largest lift threshold, 0.8, allows rules below independence. The selected values are consistent with validation favoring broader signal coverage over strict rule filtering.
3. **The signal weight varies across its full range.** The selected weights are $\gamma = 0.5, 1, 2, 4$ in 8, 2, 8, 7 runs, respectively. The smallest and largest searched values together account for 15 of 25 selections, so no single weight dominates, and the mean of 2.0 reflects this spread rather than a typical setting. Row-max normalization is chosen 16 times and log-damped row-max 9 times.

5.5 Stability Across Seeds

Seed-to-seed variation of CPARMS-WMF is small, as the sample standard deviations in Table 4 show. The largest overall standard deviation is 0.18 on Luxury Beauty, followed by about 0.11 on Pantry and 0.10 on Instruments. The one-positive-interaction group varies more, especially on Luxury Beauty and Instruments, because each user has a shorter interaction history and the evaluated subgroup is smaller.

Table 4 NDCG@10 by positive interaction group, in percent (mean $\pm$ sample standard deviation over seeds 42–46). Groups are defined by the number of positive interactions; “All” averages over all evaluated users. MostPop is deterministic and hyperparameter-free, so its standard deviation is zero by construction. The best mean per dataset and group is in bold and the second-best is underlined

Dataset	Model	All	1	2	3 ⁺
Luxury Beauty	MostPop	0.25 $\pm$ 0.00	1.09 $\pm$ 0.00	0.00 $\pm$ 0.00	0.18 $\pm$ 0.00
	Standard-WMF	4.27 $\pm$ 0.26	<u>9.53</u> $\pm$ 0.41	5.35 $\pm$ 0.71	3.32 $\pm$ 0.23
	CoFactor-WMF	4.33 $\pm$ 0.26	9.79 $\pm$ 0.41	5.31 $\pm$ 0.72	3.37 $\pm$ 0.25
	CPARMS-WMF	4.65 $\pm$ 0.18	9.24 $\pm$ 0.79	5.76 $\pm$ 0.71	3.79 $\pm$ 0.18
Industrial & Scientific	MostPop	2.58 $\pm$ 0.00	6.87 $\pm$ 0.00	4.83 $\pm$ 0.00	1.33 $\pm$ 0.00
	Standard-WMF	1.83 $\pm$ 0.11	2.71 $\pm$ 0.16	1.95 $\pm$ 0.22	1.65 $\pm$ 0.19
	CoFactor-WMF	1.86 $\pm$ 0.04	2.78 $\pm$ 0.22	1.98 $\pm$ 0.29	<u>1.68</u> $\pm$ 0.09
	CPARMS-WMF	2.89 $\pm$ 0.05	<u>6.58</u> $\pm$ 0.17	<u>4.36</u> $\pm$ 0.22	1.93 $\pm$ 0.06
Pantry	MostPop	1.44 $\pm$ 0.00	<u>2.11</u> $\pm$ 0.00	1.72 $\pm$ 0.00	1.27 $\pm$ 0.00
	Standard-WMF	<u>1.84</u> $\pm$ 0.07	1.75 $\pm$ 0.28	1.68 $\pm$ 0.09	<u>1.88</u> $\pm$ 0.05
	CoFactor-WMF	1.78 $\pm$ 0.17	1.54 $\pm$ 0.10	1.81 $\pm$ 0.33	1.82 $\pm$ 0.17
	CPARMS-WMF	1.97 $\pm$ 0.11	2.23 $\pm$ 0.16	<u>1.75</u> $\pm$ 0.19	1.97 $\pm$ 0.13
Music	MostPop	0.44 $\pm$ 0.00	0.59 $\pm$ 0.00	0.43 $\pm$ 0.00	0.41 $\pm$ 0.00
	Standard-WMF	1.72 $\pm$ 0.05	1.74 $\pm$ 0.12	2.73 $\pm$ 0.24	1.53 $\pm$ 0.05
	CoFactor-WMF	<u>1.74</u> $\pm$ 0.05	1.87 $\pm$ 0.21	2.73 $\pm$ 0.12	<u>1.53</u> $\pm$ 0.05
	CPARMS-WMF	1.83 $\pm$ 0.04	<u>1.76</u> $\pm$ 0.14	2.96 $\pm$ 0.13	1.63 $\pm$ 0.05
Instruments	MostPop	2.11 $\pm$ 0.00	4.58 $\pm$ 0.00	2.93 $\pm$ 0.00	1.56 $\pm$ 0.00
	Standard-WMF	1.95 $\pm$ 0.04	2.29 $\pm$ 0.19	2.14 $\pm$ 0.15	1.86 $\pm$ 0.05
	CoFactor-WMF	2.03 $\pm$ 0.07	2.35 $\pm$ 0.21	2.22 $\pm$ 0.09	<u>1.94</u> $\pm$ 0.10
	CPARMS-WMF	2.59 $\pm$ 0.10	<u>3.85</u> $\pm$ 0.48	3.19 $\pm$ 0.12	2.27 $\pm$ 0.08

Table 5 Overall NDCG at multiple ranking cutoffs, in percent (mean $\pm$ sample standard deviation over seeds 42–46). NDCG@10 is the primary model-selection metric. The best mean per dataset and cutoff is in bold and the second-best is underlined

Dataset	Model	NDCG@10	NDCG@20	NDCG@50	NDCG@100	NDCG@200
Luxury Beauty	MostPop	0.25 $\pm$ 0.00	1.43 $\pm$ 0.00	2.94 $\pm$ 0.00	3.49 $\pm$ 0.00	4.42 $\pm$ 0.00
	Standard-WMF	4.27 $\pm$ 0.26	4.96 $\pm$ 0.22	6.08 $\pm$ 0.21	6.88 $\pm$ 0.21	7.87 $\pm$ 0.19
	CoFactor-WMF	<u>4.33 $\pm$ 0.26</u>	<u>4.97 $\pm$ 0.26</u>	<u>6.09 $\pm$ 0.19</u>	<u>6.90 $\pm$ 0.15</u>	<u>7.97 $\pm$ 0.15</u>
	CPARMS-WMF	4.65 $\pm$ 0.18	5.43 $\pm$ 0.17	6.58 $\pm$ 0.19	7.46 $\pm$ 0.25	8.53 $\pm$ 0.20
Industrial & Scientific	MostPop	<u>2.58 $\pm$ 0.00</u>	<u>3.01 $\pm$ 0.00</u>	<u>3.81 $\pm$ 0.00</u>	<u>4.54 $\pm$ 0.00</u>	<u>5.40 $\pm$ 0.00</u>
	Standard-WMF	1.83 $\pm$ 0.11	2.35 $\pm$ 0.11	3.20 $\pm$ 0.09	4.01 $\pm$ 0.07	4.97 $\pm$ 0.10
	CoFactor-WMF	1.86 $\pm$ 0.04	2.35 $\pm$ 0.06	3.23 $\pm$ 0.14	4.02 $\pm$ 0.14	4.98 $\pm$ 0.19
	CPARMS-WMF	2.89 $\pm$ 0.05	3.47 $\pm$ 0.10	4.40 $\pm$ 0.22	5.20 $\pm$ 0.29	6.17 $\pm$ 0.37
Pantry	MostPop	1.44 $\pm$ 0.00	2.07 $\pm$ 0.00	3.20 $\pm$ 0.00	4.47 $\pm$ 0.00	<u>5.90 $\pm$ 0.00</u>
	Standard-WMF	<u>1.84 $\pm$ 0.07</u>	<u>2.37 $\pm$ 0.07</u>	<u>3.46 $\pm$ 0.08</u>	<u>4.48 $\pm$ 0.14</u>	5.79 $\pm$ 0.19
	CoFactor-WMF	1.78 $\pm$ 0.17	2.35 $\pm$ 0.14	3.39 $\pm$ 0.11	4.44 $\pm$ 0.14	5.71 $\pm$ 0.13
	CPARMS-WMF	1.97 $\pm$ 0.11	2.56 $\pm$ 0.09	3.76 $\pm$ 0.12	5.00 $\pm$ 0.21	6.42 $\pm$ 0.36
Music	MostPop	0.44 $\pm$ 0.00	0.64 $\pm$ 0.00	0.98 $\pm$ 0.00	1.51 $\pm$ 0.00	2.00 $\pm$ 0.00
	Standard-WMF	1.72 $\pm$ 0.05	2.19 $\pm$ 0.07	3.06 $\pm$ 0.07	<u>3.98 $\pm$ 0.09</u>	5.09 $\pm$ 0.10
	CoFactor-WMF	<u>1.74 $\pm$ 0.05</u>	<u>2.22 $\pm$ 0.09</u>	<u>3.06 $\pm$ 0.07</u>	<u>3.97 $\pm$ 0.09</u>	5.06 $\pm$ 0.10
	CPARMS-WMF	1.83 $\pm$ 0.04	2.31 $\pm$ 0.04	3.20 $\pm$ 0.07	4.19 $\pm$ 0.06	5.32 $\pm$ 0.03
Instruments	MostPop	<u>2.11 $\pm$ 0.00</u>	2.59 $\pm$ 0.00	3.60 $\pm$ 0.00	4.46 $\pm$ 0.00	5.42 $\pm$ 0.00
	Standard-WMF	1.95 $\pm$ 0.04	2.62 $\pm$ 0.06	3.65 $\pm$ 0.08	<u>4.62 $\pm$ 0.10</u>	<u>5.72 $\pm$ 0.14</u>
	CoFactor-WMF	2.03 $\pm$ 0.07	<u>2.64 $\pm$ 0.05</u>	<u>3.65 $\pm$ 0.08</u>	4.60 $\pm$ 0.13	5.69 $\pm$ 0.19
	CPARMS-WMF	2.59 $\pm$ 0.10	3.35 $\pm$ 0.10	4.51 $\pm$ 0.13	5.51 $\pm$ 0.18	6.60 $\pm$ 0.27

Table 6 Training time components in minutes (mean $\pm$ sample standard deviation over five seeds). “Signal” is signal-matrix construction; “ALS” is factorization optimization time

Dataset	Model	Signal	ALS training	Total
Luxury Beauty	MostPop	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
	Standard-WMF	0.000 $\pm$ 0.000	0.108 $\pm$ 0.028	0.108 $\pm$ 0.028
	CoFactor-WMF	0.000 $\pm$ 0.000	0.162 $\pm$ 0.019	0.162 $\pm$ 0.019
	CPARMS-WMF	0.010 $\pm$ 0.002	0.324 $\pm$ 0.107	0.334 $\pm$ 0.105
Industrial & Scientific	MostPop	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
	Standard-WMF	0.000 $\pm$ 0.000	0.344 $\pm$ 0.142	0.344 $\pm$ 0.142
	CoFactor-WMF	0.001 $\pm$ 0.000	0.533 $\pm$ 0.181	0.534 $\pm$ 0.181
	CPARMS-WMF	0.052 $\pm$ 0.017	2.251 $\pm$ 0.976	2.303 $\pm$ 0.991
Pantry	MostPop	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
	Standard-WMF	0.000 $\pm$ 0.000	0.329 $\pm$ 0.114	0.329 $\pm$ 0.114
	CoFactor-WMF	0.002 $\pm$ 0.000	0.629 $\pm$ 0.174	0.631 $\pm$ 0.174
	CPARMS-WMF	0.073 $\pm$ 0.029	2.379 $\pm$ 0.806	2.452 $\pm$ 0.832
Music	MostPop	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
	Standard-WMF	0.000 $\pm$ 0.000	0.523 $\pm$ 0.156	0.523 $\pm$ 0.156
	CoFactor-WMF	0.003 $\pm$ 0.000	1.145 $\pm$ 0.271	1.148 $\pm$ 0.271
	CPARMS-WMF	0.086 $\pm$ 0.051	4.219 $\pm$ 2.656	4.305 $\pm$ 2.707
Instruments	MostPop	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
	Standard-WMF	0.000 $\pm$ 0.000	0.619 $\pm$ 0.197	0.619 $\pm$ 0.197
	CoFactor-WMF	0.003 $\pm$ 0.000	1.341 $\pm$ 0.279	1.344 $\pm$ 0.279
	CPARMS-WMF	0.217 $\pm$ 0.174	8.602 $\pm$ 6.372	8.818 $\pm$ 6.542

Table 7 Selected CPARMS-WMF hyperparameter configuration for each dataset and seed (best validation NDCG@10). K : latent dimension; sweeps: ALS iterations; α : confidence boost; γ : signal weight; K_U/K_I : user/item clusters; $\sigma_{\min}/\rho_{\min}/\ell_{\min}$: minimum rule support, confidence, and lift; norm: signal normalization (row: row max; log-row: log-damped row max)

Dataset	Seed	K	λ	sweeps	α	γ	K_U	K_I	$\sigma_{\min}$	$\rho_{\min}$	$\ell_{\min}$	norm
Luxury Beauty	42	40	0.1	30	40	0.5	3	3	0	0.001	0.8	row
	43	40	0.1	20	10	0.5	5	2	0	0.0005	0.8	row
	44	40	0.0001	20	80	2	1	2	0	0	0	row
	45	20	0.1	15	80	0.5	3	3	0.0001	0.0005	0.2	row
	46	40	0.0001	15	80	2	5	5	0.0003	0.001	0.8	row
Industrial & Scientific	42	40	0.01	25	20	2	1	1	0.0003	0	0.5	row
	43	40	0.01	20	80	4	1	2	0.0003	0	0.2	log-row
	44	40	0.001	20	20	1	1	1	0.0003	0	0.5	log-row
	45	40	0.1	15	80	4	1	1	0.0001	0.001	0	row
	46	30	0.001	15	20	4	1	1	0.00005	0.0005	0.5	log-row
Pantry	42	40	0.0001	20	80	4	3	5	0.0001	0.002	0.8	row
	43	30	0.0001	20	80	4	5	5	0.0003	0	0	row
	44	20	0.01	30	80	2	1	3	0.0001	0.0005	0	log-row
	45	30	0.0001	15	20	4	3	2	0.0001	0	0.5	log-row
	46	40	0.1	20	20	2	5	3	0.0001	0.001	0.2	log-row
Music	42	40	0.1	30	40	0.5	3	3	0	0.001	0.8	row
	43	40	0.1	25	80	0.5	3	3	0.0001	0.001	0	row
	44	40	0.001	15	80	1	2	1	0.0001	0.001	0.5	row
	45	30	0.001	30	80	0.5	2	5	0.0001	0.001	0.5	log-row
	46	40	0.0001	30	40	0.5	5	5	0	0.0005	0.2	log-row
Instruments	42	30	0.01	15	80	4	1	3	0.0001	0.0005	0	row
	43	40	0.1	20	10	0.5	5	2	0	0.0005	0.8	row
	44	40	0.0001	20	80	2	1	2	0	0	0	row
	45	30	0.1	30	40	2	3	1	0	0.002	0	row
	46	30	0.1	30	20	2	3	2	0	0.0005	0.5	log-row

6 Discussion

CPARMS-WMF improves on both WMF baselines across all five datasets, with the largest gains on Industrial & Scientific and Instruments. These are also the domains in which popularity is relatively strong, suggesting that the generic component may help when sparse histories make the WMF estimates less reliable.

CoFactor-WMF is the closest baseline because both models are feature-free extensions of the same WMF objective. CoFactor-WMF slightly exceeds Standard-WMF on four of the five dataset means, whereas CPARMS-WMF exceeds CoFactor-WMF on all five means and in 24/25 runs. Under this evaluation, CPARMS-WMF outperforms the CoFactor-WMF implementation.

Performance for the one-positive-interaction group is less consistent. CPARMS-WMF generally improves on the WMF baselines but does not consistently outperform MostPop, indicating that one positive interaction may provide too little user-specific information in popularity-dominated domains.

The advantage over the WMF baselines grows as the number of positive interactions increases. At two interactions, CPARMS-WMF wins 22/25 comparisons against Standard-WMF and 23/25 against CoFactor-WMF. At three or more interactions, it has the best mean on all datasets. However, MostPop beats both WMF baselines overall on Industrial & Scientific and Instruments.

On Industrial & Scientific, where popularity is strongest and both WMF baselines trail MostPop, validation selects $K_U = 1$ in all five runs (Section 5). The generic component then becomes a popularity-like prior, and CPARMS-WMF is the only personalized model to beat MostPop. Validation selects $K_U > 1$ on Music and in four of the five Luxury Beauty runs. As n_u grows, users activate more item and liked-item-cluster tokens, increasing the personalized part of the signal (Section 3). CPARMS-WMF can thus use a popularity-like component in a popularity-dominated domain while retaining cluster- and item-level information elsewhere.

Performance shows no clear pattern across datasets. Music has the smallest CPARMS-WMF margin over Standard-WMF (+6.4%), while Industrial & Scientific has the largest (+58.4%). The strongest competitor for users with one positive interaction also varies by domain: CoFactor-WMF on Luxury Beauty and Music, while popularity remains difficult to beat for these users on Industrial & Scientific and Instruments. Pantry is the only dataset where CPARMS-WMF has the best mean for users with one positive interaction.

7 Limitations

This study has three main limitations.

1. **Popularity stays competitive at the shortest histories.** CPARMS-WMF improves over Standard-WMF and CoFactor-WMF in most comparisons for users with one positive interaction. However, MostPop has the best mean for users with one positive interaction on Industrial & Scientific and Instruments and the best mean for users with two positive interactions on Industrial & Scientific. As a result,

the method reduces the performance loss of the WMF baselines at short histories but does not always beat a non-personalized baseline.

2. **Computational cost.** CPARMS-WMF is slower than both WMF baselines, especially on the larger datasets, because the additional signal increases the work required for ALS updates.
3. **Limited base-model integration.** The main contribution of this study is the CPARMS signal, which is designed to augment a base recommender; in this work, we integrate it only with WMF. We have not evaluated whether CPARMS can be combined effectively with other architectures, such as graph-based or neural-based. The reported results therefore establish the effectiveness of CPARMS with WMF but do not yet demonstrate that it generalizes across base models.

8 Conclusion

We examined whether interaction data alone can improve personalized recommendation under sparse implicit feedback. The proposed **CPARMS-WMF** framework clusters users and items from positive feedback, mines single-antecedent association rules over user-cluster, liked-item-cluster, and item-level tokens, and reconstructs the resulting signal through an additional ALS-optimized term.

Overall, CPARMS-WMF achieves higher ranking accuracy than both WMF baselines under the evaluated protocol and outperforms MostPop at the dataset level.

The advantage weakens for users with one positive interaction, particularly relative to MostPop.

Future work should first establish a fair protocol for users with no positive interactions. A component-wise ablation could then separate the contributions of user-cluster, liked-item-cluster, and item-level antecedents. Future studies should also integrate the CPARMS signal with other base recommenders, including graph-based and neural matrix-factorization models, to determine whether its benefits extend beyond WMF. Evaluation beyond NDCG should also examine whether the gains persist for diversity, novelty, and calibration.

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Competing interests. The authors declare no competing interests.

Ethics approval. Not applicable. This study uses an existing public review dataset and does not involve participant recruitment, intervention, or animal research.

Consent to participate. Not applicable.

Consent for publication. Not applicable.

Data availability. The Amazon review datasets used in this study are publicly available from the McAuley lab data repository [16]. Section 4 fully describes the experimental configurations, including the hyperparameter search space and the evaluation protocol, to support reproducibility.

Code availability. The implementation of CPARMS-WMF and the evaluation pipeline is publicly available at <https://github.com/peerapat-t/cparms-wmf-recsys>.

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